10 Database and Eloquent ORM

Laravel 12 provides a robust database abstraction layer with support for various database systems such as MySQL, PostgreSQL, SQLite, and SQL Server. It includes migrations for schema management, seeders for populating initial data, and Eloquent ORM for interacting with databases in an object-oriented manner. This chapter will cover essential topics related to Laravel's database system.

10.1 Laravel Database Configuration

Before using Laravel's database features, you need to configure the database connection. Laravel's configuration file for database connections is located at:

```
config/database.php
```

However, the easiest way to configure your database is by updating the `.env` file. Here's an example configuration for a MySQL database:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=myapp
DB_USERNAME=root
DB_PASSWORD=secret
```

Once the database credentials are set, you can verify the connection by running the following Artisan command:

```
php artisan migrate:status
```

If Laravel connects to the database successfully, you will see a list of migration statuses.

> *Note: Laravel 12 uses SQLite as the default database. If you want to use a different database, you need to install the corresponding PHP extension and update the `.env` file with the appropriate database credentials. Make sure database drivers are enabled in your PHP configuration.*

10.2 Migrations and Seeders